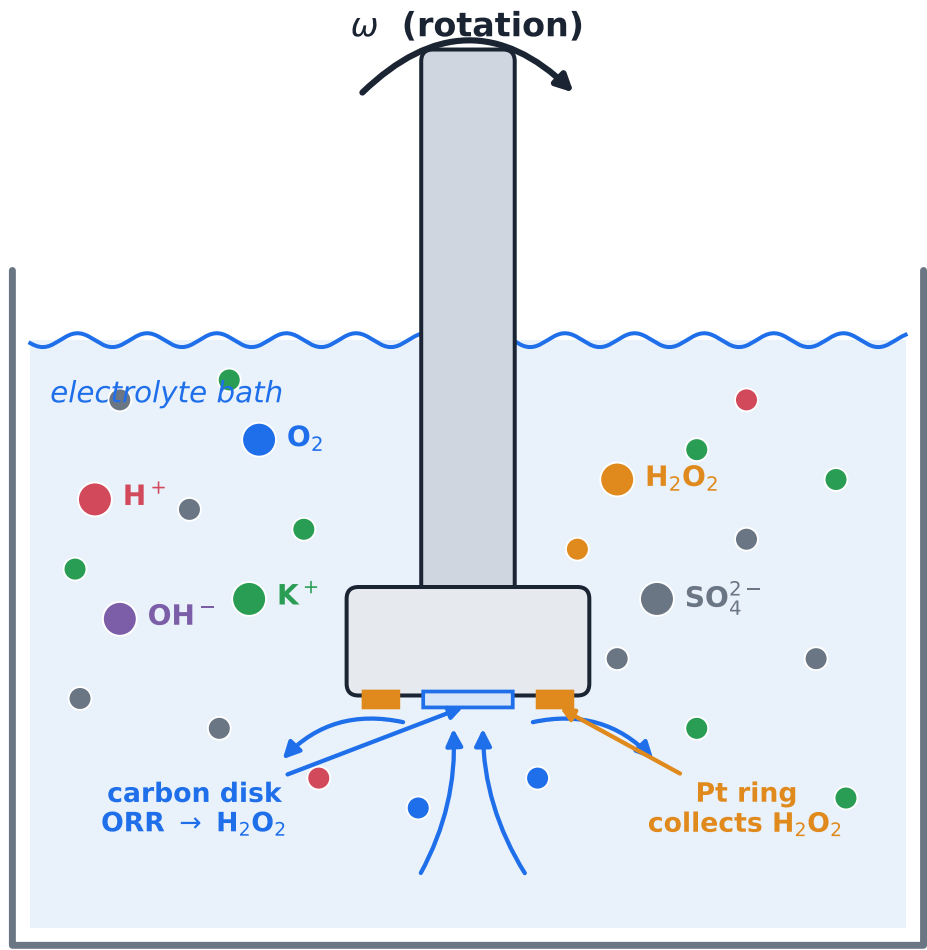


The RRDE cell: rotating ring-disk electrode in the electrolyte bath



Species in the bath

- O_2
dissolved reactant (O_2 -saturated)
- H^+
acidic medium; ORR co-reactant
- H_2O_2
 $2e^-$ product (swept to the ring)
- K^+
support cation; $\text{M}^+ = \text{Li/Na/K/Cs}$
- SO_4^{2-}
support anion ($\text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$)
- OH^-
from water / KOH (minor, acidic)
- H_2O
solvent (the bath itself)

Aqueous K_2SO_4 (M_2SO_4 , $\text{M} \in \{\text{Li,Na,K,Cs}\}$) + $\text{H}_2\text{SO}_4/\text{KOH}$ to set pH ($\approx 1\text{--}6$) · O_2 -saturated · $I \approx 0.3\text{ M}$ (Ruggiero 2022)